

TOPIC 13/60

What is Positional Encoding?

The Transformer's GPS.

How AI knows the order of words without reading left-to-right.

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THE CORE PROBLEM

Unlike older models (RNNs) that read text sequentially, Transformers read everything all at once to maximize speed.

The Side Effect:

Because it absorbs the entire sentence instantly, it inherently loses the concept of "time" or "order".

> To the raw Transformer, a sentence is just a scrambled bag of words.

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SYNTAX MATTERS



The Bag of Words

If order doesn't exist, meaning is entirely lost. Let's look at two sentences with identical words:

1. "The **dog** bit the **man**."
2. "The **man** bit the **dog**."

Without an internal concept of order, an AI cannot tell these two sentences apart. They are mathematically identical.

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THE FIRST THOUGHT

The Naive Fix

Why not just attach a simple number to every word? (1, 2, 3, 4...)

Why it fails:

Neural networks rely on incredibly small decimal numbers (between -1 and 1) to remain stable.

If a document has 50,000 words, injecting the number "50,000" into the math will completely shatter the network's stability.

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THE REAL SOLUTION

Positional Encoding

The Timestamp

Instead of raw numbers, we generate a unique, mathematical "barcode" (vector) for every single position in a sentence.

Before the word even enters the AI's brain, we **add** the word's meaning-vector to its new position-barcode. Now, the AI receives a word that knows exactly what it is *and* where it is.

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UNDER THE HOOD

Sine & Cosine

How do we create an infinite number of unique barcodes that stay perfectly bounded between -1 and 1? We use waves.

The 2017 Paper Design:

The creators of the Transformer used intersecting Sine and Cosine waves of varying frequencies to map positions.

> It acts like a highly complex clock. The fast waves map the seconds, the slow waves map the hours.

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THE BARCODE ANALOGY

POSITION #1

Wave values: [0.84, -0.12, 0.99, ...]

This specific combination of peaks and valleys forms a unique signature for the first word.

POSITION #500

Wave values: [-0.33, 0.67, 0.04, ...]

A completely different, mathematically distinct signature.

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THE MAGIC TRICK

Relative Distance

Because they are waves, the AI can easily calculate the exact relative distance between any two words.

If word A is at position 4 and word B is at position 8, the AI's math instantly recognizes they are closely linked, even if it is reading the entire 1,000-word page simultaneously.

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THE ULTIMATE SUPERPOWER

Massive Speed

By outsourcing "order" to a mathematical barcode, the AI doesn't have to read sequentially anymore.

Infinite Scale

Waves are infinite. Even if a document is 2 million words long, the positional encoding will generate a valid barcode without breaking the math.

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Positional Encoding

CONCEPT UNLOCKED



Adds a "Timestamp" to words



Uses Sine & Cosine waves



Computes relative distance



Unlocks parallel processing

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